

Beta-Two Dyson–Ornstein–Uhlenbeck Diffusion: Exact Chamber Kernel and Complete Partition Spectrum

HCS-C378 theorem package

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Abstract

We carry one trace-metric variance convention from Hermitian Ornstein–Uhlenbeck matrix diffusion to its increasingly ordered eigenvalues. The resulting beta-two system has unit noise, reciprocal-gap repulsion, linear confinement, and the ordered GUE reversible law. An exact Karlin–McGregor determinant and Vandermonde mass identity produce the conservative Doob kernel and prove noncollision from every interior start. Slater determinants of probabilists’ Hermite polynomials, divided by the Vandermonde, form a complete orthogonal basis indexed by partitions κ . Their generator eigenvalues are $-|\kappa|/2$, so level k has multiplicity $p_N(k)$ and the sharp gap is $1/2$. This also gives exact norms, the finite partition-product heat trace, and the source Fredholm determinant for every finite N . The finite artifact checks 16,602 partition labels and 12 high-precision kernels only as regression evidence. Generic random-matrix structure does not pass the Route-A arithmetic gate.

中文摘要

本文在一个自洽的迹度量方差约定中封闭贝塔二戴森奥恩斯坦乌伦贝克扩散。厄米矩阵的二阶扰动严格导出单位特征值噪声、倒间隙排斥与线性回复，有序高斯酉系综密度随之成为可逆平衡律。独立一维回复粒子的吸收行列式核作用于范德蒙德函数时产生精确次数位移，故能量修正后的杜布核守恒并给出全时间无碰撞边界。在谱侧，以概率论型埃尔米特多项式组成严格递增的斯莱特行列式，再除以范德蒙德函数，得到由分拆标记的完整正交基。完备性来自反对称埃尔米特张量基，而非有限枚举。本征值只依赖分拆大小，因而每层重数为限长分拆数，尖锐谱隙为二分之一；同一计算还给出精确范数、分拆乘积热迹和源半群的弗雷德霍姆行列式。这些对象仍是随机矩阵源系统的谱结构，没有素数、素数幂或对数素数时钟，所以严格路线甲在算术入口即停止。

Keywords: Dyson diffusion; Hermite polynomials; partition spectrum; spectral gap; heat trace; Fredholm determinant; Route A

关键词：戴森扩散；埃尔米特多项式；分拆谱；谱隙；热迹；弗雷德霍姆行列式；路线甲

1 One normalization from matrix to chamber

Let Herm_N carry the real trace inner product $\langle A, B \rangle = \text{Tr}(AB)$. Standard Brownian motion $\mathcal{B}_t$ in this Euclidean space drives

$$dH_t = d\mathcal{B}_t - \frac{1}{2}H_t dt. \quad (1)$$

The trace-orthonormal basis consists of E_{ii} and, for $i < j$,

$$\frac{E_{ij} + E_{ji}}{\sqrt{2}}, \quad \frac{i(E_{ij} - E_{ji})}{\sqrt{2}}. \quad (2)$$

Thus every diagonal matrix martingale in an instantaneous eigenbasis has quadratic variation dt , while every off-diagonal complex entry has expected squared modulus dt . This sentence fixes the factor that is often hidden in matrix-entry conventions.

Write

$$W_N = \{x \in \mathbb{R}^N : x_1 < \cdots < x_N\}, \quad h(x) = \prod_{i < j} (x_j - x_i), \quad d = \frac{N(N-1)}{2}. \quad (3)$$

Theorem 1 (Radialization in the trace convention). *If H_0 has simple spectrum, then, up to their first collision, its increasingly ordered eigenvalues solve*

$$dX_i = d\mathcal{B}_i + \sum_{j \neq i} \frac{dt}{X_i - X_j} - \frac{X_i}{2} dt, \quad (4)$$

where the B_i are independent standard real Brownian motions. The collision time is infinite. The generator on W_N is

$$\mathcal{L} = \frac{1}{2} \sum_i \partial_i^2 + \sum_i \left(\sum_{j \neq i} \frac{1}{x_i - x_j} - \frac{x_i}{2} \right) \partial_i. \quad (5)$$

Proof. For a normalized eigenvector u_i , second-order Hermitian perturbation gives

$$dX_i = \langle u_i, dH u_i \rangle + \sum_{j \neq i} \frac{|\langle u_j, d\mathcal{B} u_i \rangle|^2}{X_i - X_j}. \quad (6)$$

The covariance statement following (2) turns the first martingale term into dB_i , the second-order term into the displayed reciprocal gaps, and the matrix drift into $-X_i dt/2$. This proves the formula until the first collision. Section 2 proves that this time is infinite. $\square$

Theorem 2 (Ordered GUE reversible law). *The probability density*

$$\pi_N(x) = Z_N^{-1} e^{-|x|^2/2} h(x)^2 1_{W_N}(x), \quad Z_N = (2\pi)^{N/2} \prod_{j=0}^{N-1} j!, \quad (7)$$

is reversible for (5).

Proof. The drift is one half of the logarithmic gradient:

$$\frac{1}{2} \partial_i \log(e^{-|x|^2/2} h^2) = -\frac{x_i}{2} + \sum_{j \neq i} \frac{1}{x_i - x_j}. \quad (8)$$

Hence $\mathcal{L}f = (2\pi_N)^{-1} \nabla \cdot (\pi_N \nabla f)$, where π_N denotes its density, proving symmetry. For the normalizer, use $h = \det[\text{He}_{i-1}(x_j)]$ and scalar Hermite orthogonality in Andréief's identity. The integral over $\mathbb{R}^N$ is $N!(2\pi)^{N/2} \prod_{j=0}^{N-1} j!$. The integrand is symmetric, so the ordered chamber contributes $1/N!$ of that value. The invariant matrix density is proportional to $e^{-\text{Tr } H^2/2}$; Weyl integration gives the same ordered density. $\square$

The $N = 1$ face is ordinary scalar Ornstein–Uhlenbeck diffusion. No large- N passage or entrance from a multiple eigenvalue is included.

2 Killed determinant and conservative Doob transform

The independent scalar generator is

$$\mathcal{L}_0 = \frac{1}{2} \sum_i (\partial_i^2 - x_i \partial_i). \quad (9)$$

For $t > 0$, put $r = e^{-t/2}$ and $\sigma_t^2 = 1 - r^2$. Its Lebesgue density is

$$p_t(a, b) = \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left[-\frac{(b - ra)^2}{2\sigma_t^2}\right]. \quad (10)$$

Lemma 1 (Vandermonde energy). *The polynomial h is harmonic and homogeneous of degree d . Consequently,*

$$\mathcal{L}_0 h = -\frac{d}{2} h. \quad (11)$$

Proof. The permutation-invariant Laplacian sends the alternating polynomial h to an alternating polynomial of degree $d - 2$. Every nonzero alternating polynomial is divisible by h , which already has degree d ; hence $\Delta h = 0$. Euler's homogeneous identity gives $x \cdot \nabla h = dh$. Substitution in (9) proves (11). $\square$

Theorem 3 (Exact chamber kernel and boundary). *The independent process killed at its first collision has density*

$$q_t(x, y) = \det[p_t(x_i, y_j)]_{i,j=1}^N. \quad (12)$$

Its Vandermonde mass is

$$\int_{W_N} q_t(x, y) h(y) dy = r^d h(x). \quad (13)$$

Therefore

$$k_t(x, y) = r^{-d} \frac{h(y)}{h(x)} q_t(x, y) \quad (14)$$

is a conservative Markov density with generator (5). Starting from any $x \in W_N$, its continuous diffusion never reaches a collision wall in finite time.

Proof. The reflection principle pairs every wall-hitting independent path with the path reflected after its first hit. The signed permutation sum that remains is the Karlin–McGregor determinant (12).

Both $q_t(x, \cdot)$ and h are alternating. Their product is symmetric, so the chamber integral in (13) is $1/N!$ of the full-space integral. Write $h(y) = \det[y_j^{i-1}]$. Andréief's identity gives

$$\int_{\mathbb{R}^N} q_t(x, y) h(y) dy = N! \det \left[\int_{\mathbb{R}} p_t(x_i, y) y^{j-1} dy \right]_{i,j=1}^N. \quad (15)$$

The $(j-1)$ st moment of $N(rx_i, 1-r^2)$ is a degree- $(j-1)$ polynomial in x_i with leading coefficient r^{j-1} . Lower-degree column operations reduce the determinant in (15) to $r^{0+\dots+(N-1)} h(x) = r^d h(x)$. This proves (13).

Equation (13) makes (14) conservative. On smooth compactly supported functions,

$$h^{-1} \mathcal{L}_0(hf) + \frac{d}{2} f = \mathcal{L}_0 f + \nabla \log h \cdot \nabla f = \mathcal{L} f, \quad (16)$$

because $\partial_i \log h = \sum_{j \neq i} (x_i - x_j)^{-1}$. The minimal transformed process has mass one at every finite time. A boundary hit by a continuous chamber path would be its killing time, so conservativity excludes such a hit. Local uniqueness before collision and the standard electrostatic-repulsion theorem identify the transform with the unique global strong solution of (4). $\square$

Scalar detailed balance in (10) passes through the determinant; the two Vandermonde factors in (14) then prove $\pi_N(x) k_t(x, y) = \pi_N(y) k_t(y, x)$ directly. This independently checks the generator calculation.

3 Complete symmetric-Hermite spectrum

Let $\kappa = (\kappa_1 \geq \dots \geq \kappa_N \geq 0)$ be a partition and define

$$m_i = \kappa_{N+1-i} + i - 1. \quad (17)$$

Then $m_{i+1} - m_i = 1 + \kappa_{N-i} - \kappa_{N+1-i} \geq 1$ and $\sum_i m_i = d + |\kappa|$. With monic probabilists' Hermite, set

$$D_m(x) = \det[\text{He}_{m_i}(x_j)]_{i,j=1}^N, \quad \Phi_\kappa(x) = \frac{D_m(x)}{h(x)}. \quad (18)$$

The quotient is a symmetric polynomial because every alternating polynomial is divisible by h .

Theorem 4 (Complete partition spectrum and exact norms). *The family $\{\Phi_\kappa\}$ is a complete orthogonal basis of $L^2(W_N, \pi_N)$ and*

$$\mathcal{L} \Phi_\kappa = -\frac{|\kappa|}{2} \Phi_\kappa, \quad (19)$$

$$\|\Phi_\kappa\|_{L^2(\pi_N)}^2 = \frac{\prod_{i=1}^N m_i!}{\prod_{j=0}^{N-1} j!}. \quad (20)$$

Thus $-k/2$ has multiplicity $p_N(k)$, the number of partitions of k with at most N parts.

Proof. The scalar equation

$$\frac{1}{2}(\text{He}_n'' - x \text{He}_n') = -\frac{n}{2} \text{He}_n \quad (21)$$

and determinant expansion give $\mathcal{L}_0 D_m = -(\sum_i m_i) D_m / 2$. Apply (16) to $D_m = h \Phi_\kappa$ and subtract the filled-ground energy $d/2$ to obtain (19).

Scalar Hermite form a complete Gaussian $L^2(\mathbb{R})$ basis. Their antisymmetrized tensor products D_m , indexed by strictly increasing m , therefore form a complete basis of the antisymmetric Gaussian sector on $\mathbb{R}^N$. Multiplication by h , followed by antisymmetric extension from W_N , identifies this sector with $L^2(W_N, \pi_N)$ up to a fixed scalar. Division by h transfers the complete Slater basis to (18); this is the step that rules out missing nonpolynomial L^2 spectrum.

Finally, Andréief and scalar Hermite orthogonality give

$$\int_{\mathbb{R}^N} D_m D_n e^{-|x|^2/2} dx = \delta_{mn} N! (2\pi)^{N/2} \prod_i m_i!. \quad (22)$$

Divide by $N!$ for the chamber and by Z_N to obtain (20). Equation (17) bijects partitions with strict indices above $(0, 1, \dots, N-1)$, proving the multiplicity. $\square$

Corollary 1 (Sharp gap, trace, and source determinant). For $\int f d\pi_N = 0$,

$$\|P_t f\|_2 \leq e^{-t/2} \|f\|_2, \quad \text{Var}_{\pi_N}(f) \leq \int_{W_N} |\nabla f|^2 d\pi_N. \quad (23)$$

Both constants are sharp. For $t > 0$,

$$\text{Tr } P_t = \prod_{j=1}^N (1 - e^{-jt/2})^{-1}, \quad (24)$$

$$\det(I - zP_t) = \prod_{k=0}^{\infty} (1 - ze^{-kt/2})^{p_N(k)}. \quad (25)$$

Proof. Theorem 4 has a simple zero level and first nonzero decay exponent $1/2$. The function $x_1 + \cdots + x_N$ lies in that first eigenspace, proving sharpness. The partition generating function $\sum_{\kappa} q^{|\kappa|} = \prod_{j=1}^N (1 - q^j)^{-1}$ with $q = e^{-t/2}$ proves (24). The finite value of the trace makes P_t trace class, and its eigenvalue product gives (25). This determinant belongs only to the source semigroup. $\square$

4 Oscillator transform, exact receipt, and Route A

The equilibrium square root without its constant is $\psi_0 = e^{-|x|^2/4}h$. Direct differentiation, using $\Delta h = 0$ and $x \cdot \nabla h = dh$, gives

$$-\psi_0 \mathcal{L} \psi_0^{-1} = -\frac{1}{2} \Delta + \frac{|x|^2}{8} - \frac{N}{4} - \frac{d}{2}. \quad (26)$$

This acts on the antisymmetric Dirichlet sector; the inverse-square Calogero term cancels at beta two. It explains the filled Slater sea but is only a source-local operator equivalence.

The canonical receipt records 16 matrix dimensions, 1,040 level multiplicities, all 16,602 partitions through $N = 8$ and degree 24, and 12 high-precision kernels. An independent checker reconstructs every exact row and uses a Leibniz determinant. SymPy performs 350 exact identities. These finite checks are regression evidence, not the proof of the all- N theorem.

Dyson owns the eigenvalue Brownian-motion lineage [1]; Karlin–McGregor and Grabiner own the determinant/reflection lineage [2, 4]; Cépa–Lépingle supply strong noncollision context [3]; and Baker–Forrester place generalized Hermites in the Calogero–Sutherland setting [5]. The nearest registry owners are a deterministic Calogero–Moser scattering system, a scalar Jacobi diffusion, a hypoelliptic Kramers process, and finite discrete killed walkers. None is the continuous conservative eigenvalue diffusion closed here.

The strict assessment is

$$(A0_{\text{fail}}, A1_{\text{fail}}, A2_{\text{fail}}, A3_{\text{fail}}, A4_{\text{formal hint}}), \quad \text{overall: Route A rejected.} \quad (27)$$

The scope is NO_BAD_EULER_OR_ROOT_NUMBER. Generic GUE statistics are not arithmetic origin. We assert no target arithmetic local data, Euler factor, root number, automorphy, target divisor or counting law, target functional equation, target-zero match, Hilbert–Pólya operator, or Route B. In particular, (26) is not promoted to a target spectral realization.

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